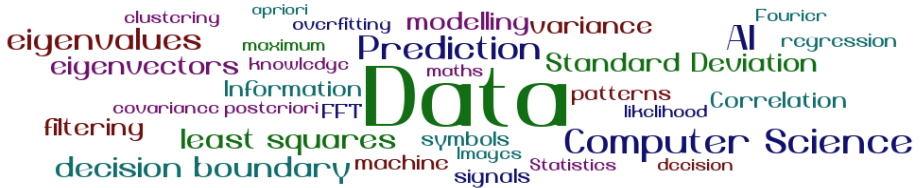


COMS20017 – Algorithms & Data

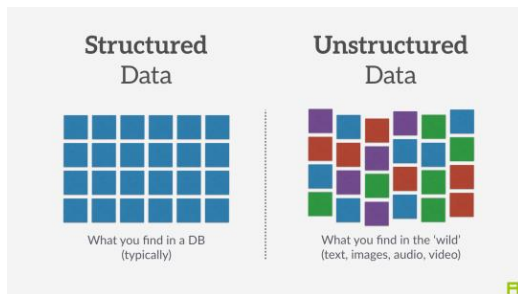


September 2026
Majid Mirmehdi

Lecture MM-01

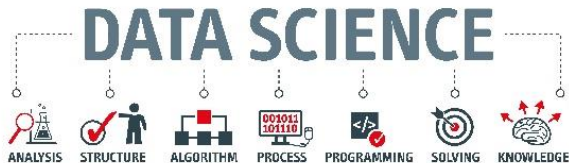
What is Data?

- Data comes in many forms, e.g. text, symbols, patterns and signals!
- Data: *Structured and Unstructured*
 - Numeric (measurements, finance spreadsheets, ...)
 - Textual (emails, social media, web pages, medical records, ...)
 - Visual (images, video, graphics, animations)
 - Auditory (speech, audio)
 - Signals (GPS signals, accelerometer, heart rate, ...)
 - Many others...

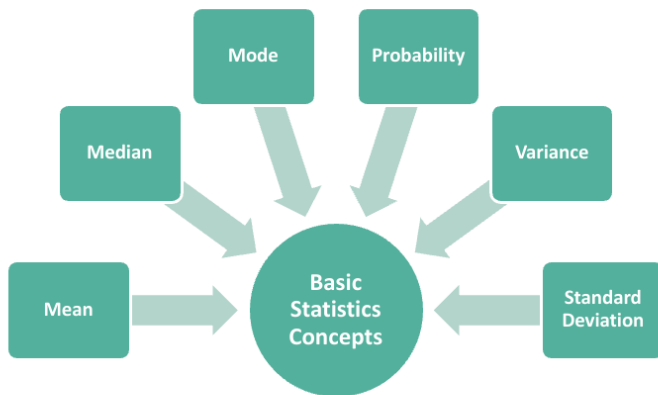


This Unit

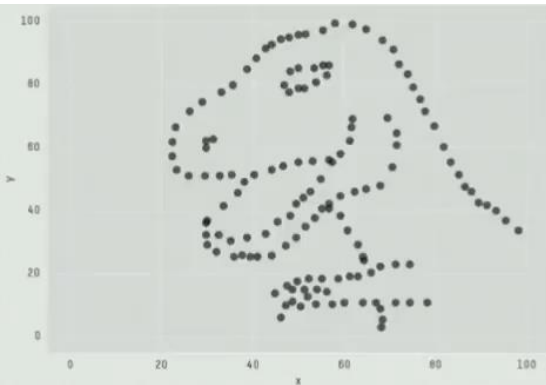
- This unit is about doing things with data... *but not*
 - storing, shuffling, searching (Algorithms)
 - sending (Computer Systems)
 - compressing or encrypting (Cryptography)
- This unit is about:
 - extracting knowledge from data
 - generating data and making predictions
 - making decisions based on data
 - Often referred to as:



Basic Statistics Concepts



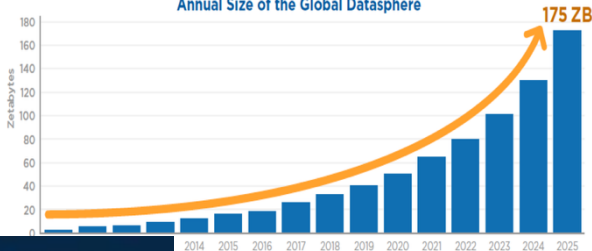
Same Basic Stats, Different Data!



X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526

Amount of Data!

Annual Size of the Global Datasphere



Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018

Data We Create Online in 60 Seconds



Data is the new Oil

Chart of the Week

THE LARGEST COMPANIES BY MARKET CAP

The oil barons have been replaced by the whiz kids of Silicon Valley



Top 5 Publicly Traded Companies (by Market Cap)



Tech



Other



visualcapitalist.com



Largest Companies in The UNITED STATES and CHINA

United States



By Market Cap

As of June 26, 2025

China



Source: Companies Market Cap
www.rankingroyals.com



RankingRoyals



Example Job Positions Involving Data

Data Analyst

- + Data retrieval
- + Spot trends and patterns
- + Visualise and report to others

Data Engineer

- + Design and maintain data management systems
- + Make data accessible to others

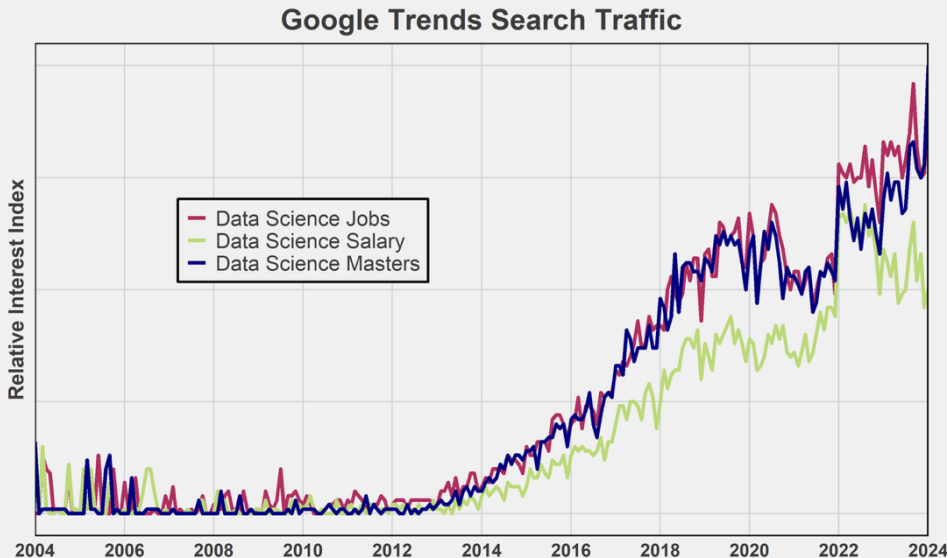
- + Use ML techniques to derive insights
- + Make predictions on products, assets, etc. based on past data

Data Scientist

- + Design and implement ML methods
- + Extend existing ML frameworks and libraries

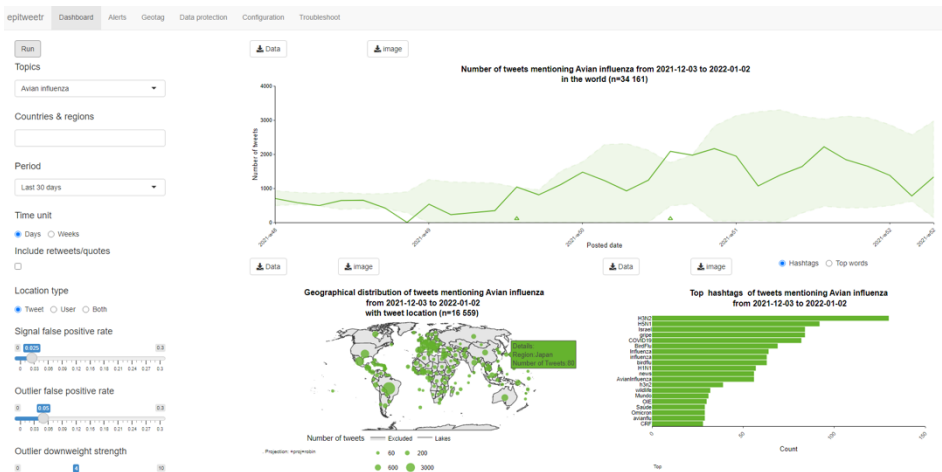
ML Engineer

Data Science & Analytics



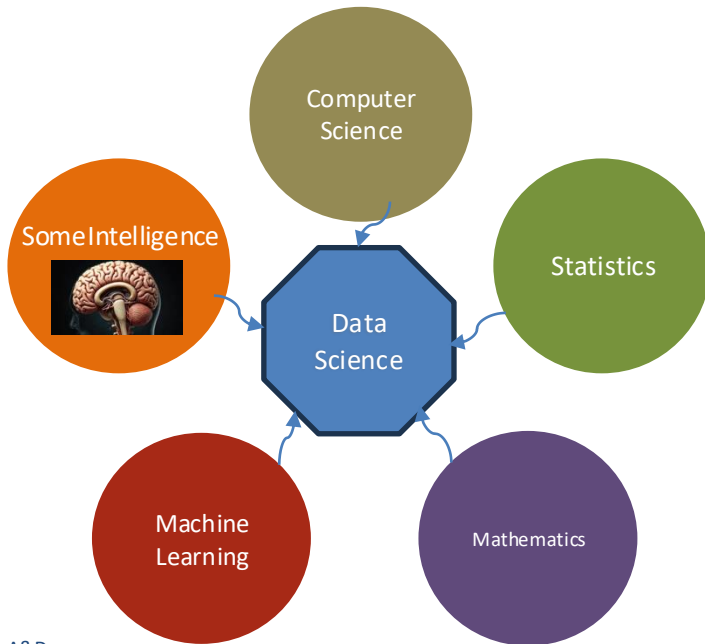
It's not about the data – it's about the science

Example: Tracking and predicting [disease,mortality,floods,fires,fun etc.] by Twitter!



<https://www.dicardiology.com/artide/understanding-how-big-data-will-change-health-care>

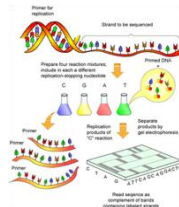
It's not about the data – it's about the science



This Unit

Why is it important for Computer Science?

- Fundamental to many related areas:
 - Artificial Intelligence, Machine Learning, Deep Learning
 - Image Processing and Pattern Recognition
 - Graphics, Animation and Virtual Reality
 - Computer Vision and Robotics
 - Speech and Audio Processing
 - With growing applications in: neuroscience, literature, agriculture, etc.
- Hence, preparation for units in years 3 and 4.



<https://www.bris.ac.uk/unit-programme-catalogue/UnitDetails.jsa?unitCode=COMS20017>

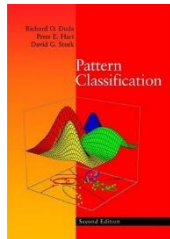
Ex1. A Fish Problem



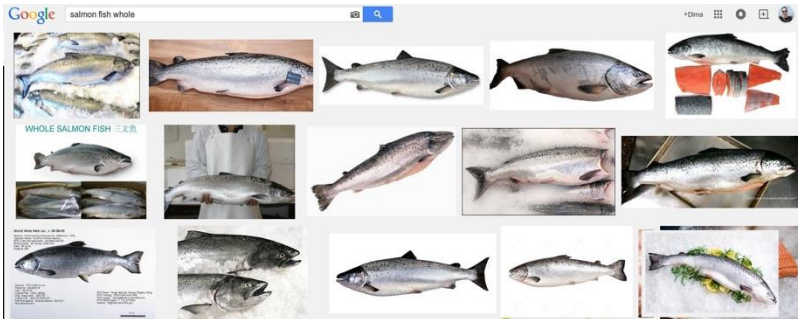
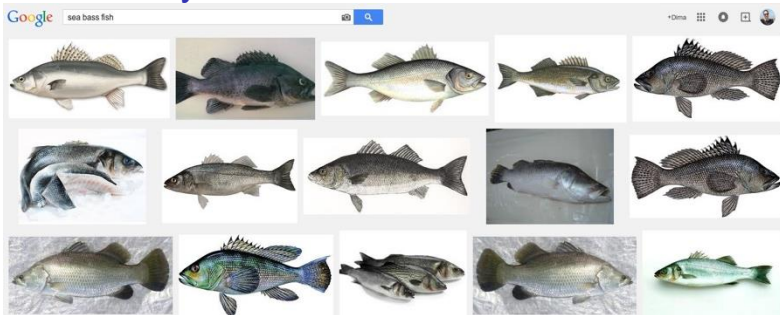
Data: images of fish

Aim: distinguish between sea bass and salmon

From: Pattern Classification by *Duda, Hart and Stork*,
2nd Edition, Wiley Interscience



Ex1. A Fishy Problem

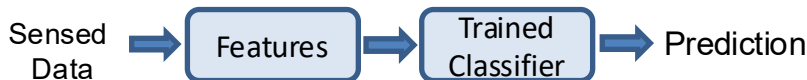


Features

They are the intrinsic traits, properties, or characteristics that tell one data/pattern/object apart from another.

Feature extraction and representation allows:

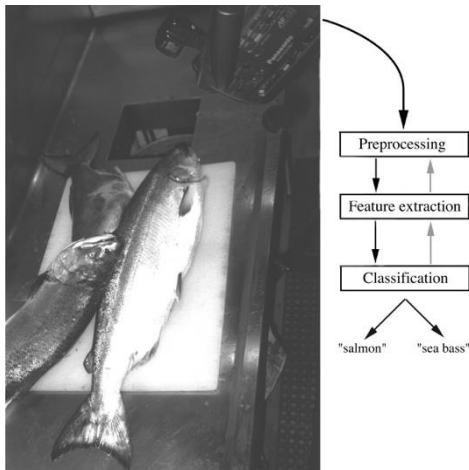
- Data reduction and abstraction
- Focus on relevant, distinguishing parts of data



Fishing for a Solution

Steps:

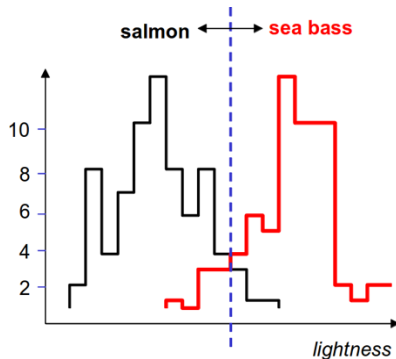
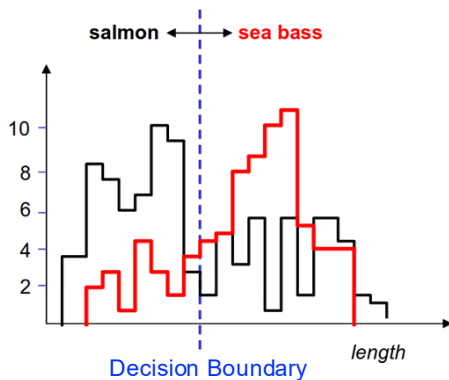
1. Pre-processing e.g. Rotate and align, Segment fish from background
2. Feature Selection e.g. Measure length
3. Classification e.g. Find a threshold



Fishing for a Solution

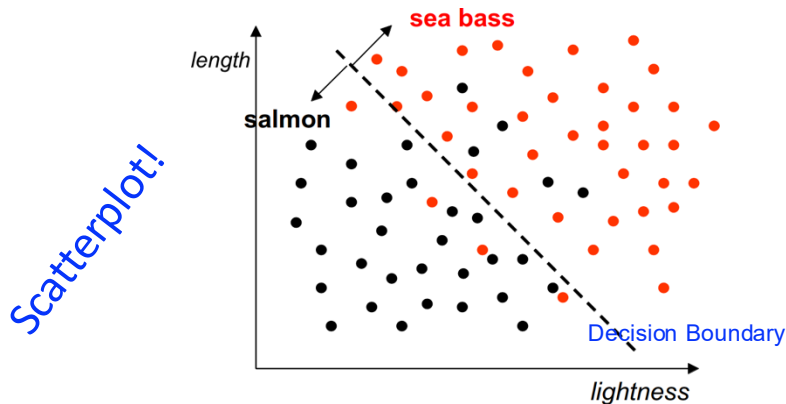
Steps:

1. Pre-processing e.g. Rotate and align, Segment fish from background
2. Feature Selection e.g. Measure length or lightness
3. Classification e.g. Find a threshold



Fishing for a Solution

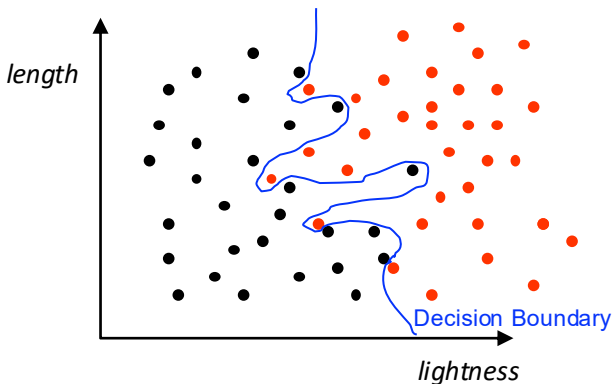
Multiple features could be selected, resulting in a multi-dimensional feature vector.



$$\text{Fish} \rightarrow \mathbf{x} = \{x_1, x_2\}$$

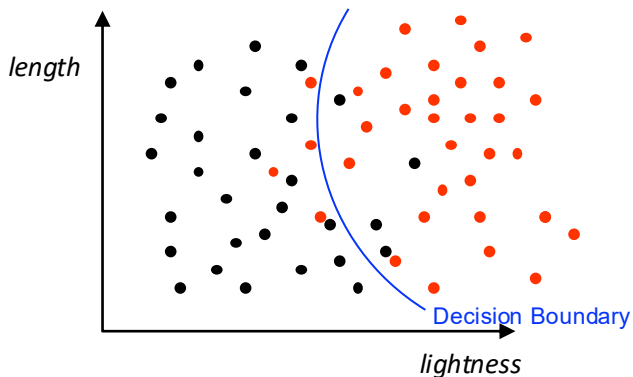
Fishing for a Solution

Complex decision model

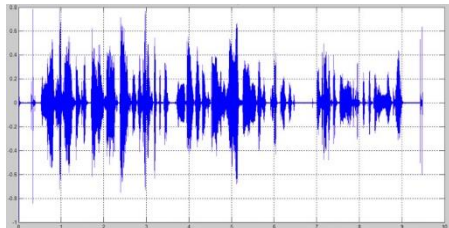


Fishing for a Solution

Optimal trade-off between performance and generalization



Ex2. Speech Recognition



Data: Analogue speech signals (time-series numerical data)

Aim: Convert audio into text (e.g. Alexa/Siri...)

1. Pre-processing Digitisation
2. Feature Selection Wave amplitude, frequencies
3. Inference Hidden Markov Models (Viterbi algorithm) or Deep learning

Ex3. Spam Filter

Data: Texts of emails

Aim: Determine whether the email is spam



1. Pre-processing - **Normalise words** (e.g. remove punctuation, find word roots)
2. Feature Selection - **Presence of words**
3. Classification - **Naive Bayes classifier**

Select subset of words w_i and determine $P(w_i | spam)$ and $P(w_i | \neg spam)$ from frequencies in training data.

For an Email that contains $w_1, w_2, \dots, w_n$ of the subset of words, assume

$$P(email | spam) = P(w_1 | spam)P(w_2 | spam) \dots P(w_n | spam) \quad (1)$$

and

$$P(email | \neg spam) = P(w_1 | \neg spam)P(w_2 | \neg spam) \dots P(w_n | \neg spam) \quad (2)$$

A new Email is spam if

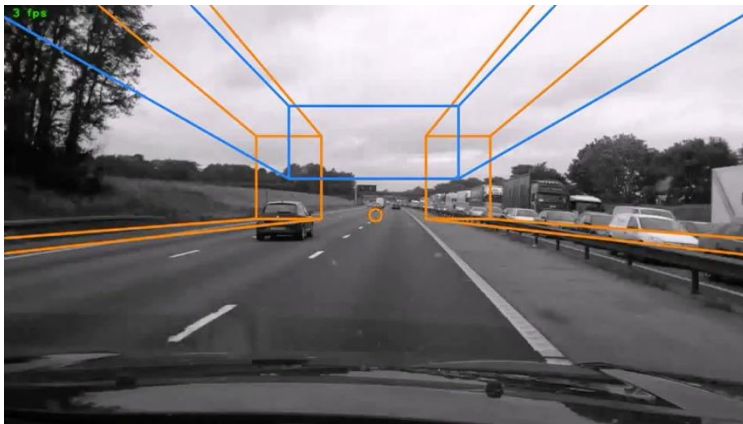
$$\underline{P(email | spam)} > \underline{P(email | \neg spam)} \quad (3)$$

Ex4.1 – Towards Autonomous Driving

Data: Video

Aim: Determine knowledge from the road or inside the vehicle

1. Pre-processing (Detect vanishing point)
2. Feature Selection (Use constraints to reduce number and dimensionality)
3. Recognition (Perspective transformations and OCR)



Ex4.2 – Towards Autonomous Driving

1. Pre-processing (Detect vanishing point)
2. Feature Selection (Straight lines)
3. Model Building (Detecting, predicting, decision making)



Ex4.3 – Towards Autonomous Driving

1. Pre-processing (Detect vanishing point)
2. Feature Selection (MSERs, Histogram of Gradients)
3. Classification (Support Vector Machines)



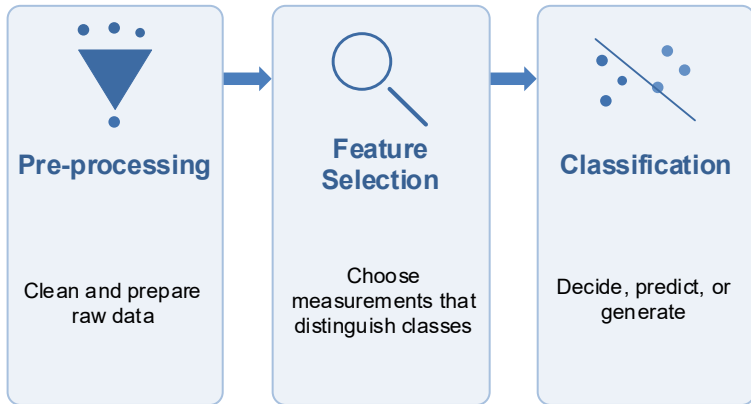
Ex4.4 – Towards Autonomous Driving

1. Pre-processing (Background subtraction)
2. Feature Selection (hand shapes)
3. Classification (Random Forest classifier)



Same Recipe

Every example we saw follows the same three steps



The Data unit traces the recipe (more or less)!

Key Terms to Know

Some vocabulary you'll meet throughout this unit:



Feature

Measured
property
of data



Label

Known class
attached to
training data



Model

Classifier, i.e.
learned
mapping from
features to
predictions



**Training /
Test**

Data used to
learn vs. to
evaluate



Generalisation

Performing well
on unseen data

Supervised Learning

Common Pitfalls

Real data and models rarely behave perfectly!



Noisy or Missing Data

Errors, gaps, and unreliable readings



Imbalanced or Biased Samples

Some classes are over- or under-represented



Curse of Dimensionality

Too many features, too little data to fill the space



Overfitting

Memorising the training data instead of generalising

Much of this unit is about recognising and managing these issues.

Data Ethics

Extracting knowledge from data brings responsibility



Bias

Models can inherit and amplify bias in training data



Privacy

Personal data must be collected and used responsibly



Fairness

Predictions can affect real people's lives



Transparency

Explainability matters, not just accuracy

Good data science asks not just can we, but should we?

COMS20017 - Data

Steps:

1. Pre-processing
 2. Feature Selection
 3. Modelling & Classification
- } [Part 1] → Majid Mirmehdi (~50%)
- [Part 2] → Alin Achim (~50%)

Part 1 – supported with Problem Sheets

Part 2 – supported with Problem Sheets and Labs



COMS20017 - Data

Unit pages: https://majidmirmehdi.github.io/COMS20017_DATA_26-27/

Lectures

Mondays 3pm

PHYS BLDG G42 POWELL

Tuesdays 4pm

ST. MICHAEL'S HILL

Labs starting Week 5

Thursdays 1 – 2pm [by timetable]: Group 1

Thursdays 2 – 3pm [by timetable]: Group 2

Lab Environment [Jupyter + Python]

TA support in unit's Teams group

Lectures, Problemsheets and Labs are all
essential for learning unit content!

Very Welcome to Ask Questions...

You should **use the unit's Teams channel** for raising queries on whatever aspects of the COMS20017 Data unit!

Queries will normally only be answered via email or via personal Teams messages, **IF it is a personal question that cannot be shared.**

Please post your query on the unit Teams channel for the benefit of others who may have the same query.

Next lecture



Analog Signal



Digital Signal

- **Data acquisition**
- **Data characteristics: distance measures**
- Data characteristics: summary statistics [*reminder*]
- Data normalisation and outliers